Ege C. Kaya

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Research Profile

PhD candidate in reinforcement learning and optimization, graduating May 2027 and available for full-time work in Summer 2027. Develops new formalisms, algorithms, and finite-sample theory for distributional, multi-objective, risk-sensitive, and average-reward RL, as well as optimization methods for LLM post-training. Research combines mathematical analysis with implementation and empirical evaluation in tabular and neural environments.

Education

Purdue University, West Lafayette, IN — August 2022–May 2027 (expected)
PhD Candidate, Electrical and Computer Engineering — Advisor: Prof. Abolfazl Hashemi
Thesis: *Uncertainty, Risk, Robustness and Multi-Objective Reasoning in Sequential Decision-Making*
MS, Electrical and Computer Engineering — Awarded August 2026
Boğaziçi University, Istanbul, Turkey — September 2018–June 2022
BSc, Computer Engineering and BSc, Mathematics (double major) — High honors in both majors

Research Experience

Purdue University — West Lafayette, IN
Graduate Research Assistant, MINDS Research Group — *August 2022–Present*

- Developed **LOO-ROLL** for LLM post-training, with finite-rank theory for EGGROLL's update field and estimator variance. At matched training time, LOO-ROLL improved Qwen3-14B mean accuracy over EGGROLL by 7.85 percentage points on GSM8K and 8.08 percentage points on MATH-500.
- Introduced **joint MDPs**, a formalism for coupled-dynamics environments that preserves cross-action counterfactual dependence. Developed dynamic-programming and incremental algorithms with convergence guarantees and evaluated them in tabular and neural environments.
- Developed finite-iteration and structural theory for categorical distributional RL, covering asynchronous TD under i.i.d. and Markovian sampling and quotient-categorical representations for average-reward problems.
- Established convergence and oracle-complexity results for stochastic approximation and optimization, including non-expansive fixed-point iteration with Markovian data and non-convex optimization under unbounded variance.

Dolby Laboratories — San Francisco, CA
PhD Research Intern — *May 2025–August 2025*

- Conceived and built **ORDO**, a first-order policy-optimization method comparing multivariate return distributions through stochastic dominance, during a three-month project on experience delivery and media personalization.
- Continued the work after the internship by establishing approximate non-dominatedness guarantees and developing stochastic estimators. Published the resulting research at UAI 2026.

Additional Experience

Nokia — Istanbul, Turkey
Data Analytics and Automation Intern — *July–September 2021*

- Developed customer network-data processing and visualization tools using the Elasticsearch-Logstash-Grafana stack.

Scientific and Technological Research Council of Turkey (TÜBİTAK) — Istanbul, Turkey
STAR Research Intern — *February–August 2021*

- Studied deep-learning methods for single-image super-resolution.

Peer-Reviewed Publications

1. **Kaya, E.C.**, Pourghani, A., Gupta, V., Hashemi, A. "Quotient-Categorical Representations for Bellman-Compatible Average-Reward Distributional Reinforcement Learning." **NeurIPS**, 2026. Preprint: arXiv:2605.11289.
2. **Kaya, E.C.**, Ghasemi, M., Hashemi, A. "Joint MDPs and Reinforcement Learning in Coupled-Dynamics Environments." **Uncertainty in Artificial Intelligence (UAI, Oral)**, 2026. Oral presentation, top 2.2% of 1,087 submissions.
3. **Kaya, E.C.**, Kaheman, K., Cloud, J.M., Hashemi, A. "Stochastic Dominance Driven First-Order Policy Optimization for Multi-Objective Reinforcement Learning." **UAI**, 2026.
4. **Kaya, E.C.**, Hibbard, M., Tanaka, T., Topcu, U., Hashemi, A. "Randomized Greedy Methods for Weak Submodular Sensor Selection with Robustness Considerations." **Automatica**, 171, 2025.
5. **Kaya, E.C.**, Hashemi, A. "Localized Distributional Robustness in Submodular Multi-Task Subset Selection." **IEEE Transactions on Signal Processing**, 2024.

6. Jiménez, A.C.C., **Kaya, E.C.**, Ye, L., Hashemi, A. "Equitable Client Selection in Federated Learning via Truncated Submodular Maximization." **IEEE Conference on Decision and Control (CDC)**, 2024.
7. **Kaya, E.C.**, Sahin, M.B., Hashemi, A. "Communication-Efficient Zeroth-Order Distributed Online Optimization: Algorithm, Theory, and Applications." **IEEE Access**, 2023.
8. **Kaya, E.C.**, Hashemi, A. "Relative Entropy Regularization for Robust Submodular Multi-Task Subset Selection." **Allerton Conference**, 2023.
9. **Kaya, E.C.**, Sahin, M.B., Hashemi, A. "Communication-Constrained Exchange of Zeroth-Order Information with Application to Collaborative Target Tracking." **ICASSP**, 2023.
10. Jiménez, A.C.C., **Kaya, E.C.**, Hashemi, A. "High Probability Guarantees for Submodular Maximization via Boosted Stochastic Greedy." **Asilomar Conference**, 2023.

Workshop Publications

1. **Kaya, E.C.**, Ghasemi, M., Hashemi, A. "Beyond Marginals: Capturing Correlated Returns through Joint Distributional Reinforcement Learning." **NeurIPS Workshop on Aligning Reinforcement Learning Experimentalists and Theorists (ARLET)**, 2025.

Manuscripts and Preprints

1. **Kaya, E.C.**, Hashemi, A. "EGGROLL, Unrolled: Understanding and Improving Low-Rank Evolution Strategies at Scale." Preprint available: arXiv:2609.10980.
2. Sahin, M.B., **Kaya, E.C.**, Fazla, A., Sharif, B., Hashemi, A. "Robust Langevin Sampling under State-Dependent Heavy-Tailed Noise with Applications to LLM Fine-Tuning."
3. Fazla, A., Upadhyay, A., **Kaya, E.C.**, Sahin, M.B., Hashemi, A. "Why Does Adaptive Batching Help LLM Pretraining? A Perspective from Unbounded Variance."
4. Fazla, A.*, **Kaya, E.C.***, Upadhyay, A.*, Hashemi, A. "Lower Bounds and Proximally Anchored SGD for Non-Convex Minimization Under Unbounded Variance." *Equal contribution. Preprint available: arXiv:2604.16620.
5. **Kaya, E.C.**, Pourghani, A., Ghasemi, M., Gupta, V., Hashemi, A. "Learning to Control Coupled-Dynamics Environments with Joint Markov Decision Processes." Preprint available: arXiv:2608.22765.
6. **Kaya, E.C.**, Fazla, A., Sahin, M.B., Hashemi, A. "A Banach-Space Theory of Markovian Halpern Iteration for Non-Expansive Maps." Preprint available: arXiv:2608.15966.
7. **Kaya, E.C.**, Hashemi, A. "A Finite-Iteration Theory for Asynchronous Categorical Distributional Temporal-Difference Learning." Preprint available: arXiv:2605.06866.

Teaching Experience

Purdue University West Lafayette, IN
Graduate Teaching Assistant *August 2024–May 2025*
- ECE 600: Random Variables and Signals. ECE 369: Discrete Mathematics for Computer Engineering.

Boğaziçi University Istanbul, Turkey
Undergraduate Teaching Assistant *January–June 2022*
- CMPE 160: Object-Oriented Programming.

Technical Skills

Research: reinforcement learning, distributional RL, multi-objective and risk-sensitive RL, LLM post-training, evolution strategies, stochastic approximation, optimization, probabilistic modeling, Monte Carlo methods, submodular optimization
Programming and tools: Python, PyTorch, JAX, vLLM, Ray, MATLAB, Git, Docker

Honors and Service

NeurIPS 2026 Top Reviewer.
Bilsland Dissertation Fellowship, Purdue University. Competitive final-year doctoral fellowship.
Cahit Arf Grant, Republic of Turkey. National scholarship awarded based on top-100 placement in the university entrance examination.
Reviewer. AISTATS 2024, AAAI 2025, UAI 2025–2026, NeurIPS 2026, ICLR 2027, and NeurIPS ARLET Workshop 2025.
Languages: Turkish (native), English (proficient), French (advanced).